The price of remembering cheaply: *O(n)-exact vs O(1)-lossy memory*

Topic: Key–Value Caching, Limitations, and Alternate Approaches · Artifact: `mbahc25088.github.io/IITKGP` · Repo: `github.com/mbahc25088/IITKGP`

OUR FALSIFIABLE CLAIM

Transformer KV-cache grows one slot per token — O(n), always exact. Fixed recurrent state stays O(1) but silently forgets when writes collide. Flood 500 pairs into 16 slots: if recall stays perfect, we lose.

Why now. Long context made exactness unaffordable: every new token attends over all old tokens. Linear attention, SSMs, and recurrent states collapse the cache to a fixed matrix updated incrementally — O(T) compute, O(1) memory — at the cost of interference between non-orthogonal keys.

WHAT CHANGES, TECHNICALLY

Per-token slots become an additive state (BDH-CQ notes the case where state accumulates additively per demonstration). Our toy: cabinet = `Map` per pair; whiteboard = `slot = hash % N`, last-writer-wins. At 120/32 collisions already bite; 500→16 craters recall while KV stays 100% and its slot-line climbs against the flat O(1) line. Clicking a yellow cell names the thieves — the gap *is* the lesson.

LANDSCAPE — WHO PAYS WHAT

Family	Memory	Recall	Bill	Signal
Transformer KV	O(n)	perfect	$O(T^2)$, OOMs	gold standard
Mamba SSM ’24	O(1) gated	lossy far	cheap	throughput
HOLA ’26	O(1)+cache	recovered	cheap+cache	surprise β‖e‖
BDH / CQ	**O(1) synaptic ρ, S_t**	sags w/ load	lowest $/task shown	memory=reasoning

BDH, PLAINLY

BDH (Dragon Hatchling) is the O(1)-by-design case: synaptic state via Hebbian writes, not drawers — $\rho_{t-1}=\Sigma\ v^* x^T U^{t-\tau}$ (Eq.5; §3.2, Hebbian 6–8). BDH-GPU = ReLU low-rank + linear attention, **not** a Mamba SSM. **BDH-CQ** is the sibling Pareto point: 150M, 29.5% pass@2 ARC-AGI-1 @ ~$0.0007/task, latent recurrence with no chain-of-thought.

Evidence, labeled. 134M BDH: 95%@32K → 82%@128K BABILong QA1–QA5 (pathway.com; vendor-reported, pending contamination/independent/leaderboard review). Same sag as the toy, at scale.

Cost evidence. BDH-CQ ~11× cheaper/task than GPT-5.6 Luna Low (34.2% vs 29.5%) in Pathway’s table — one self-reported benchmark, weights unreleased; Pareto claim, not raw SOTA.

LIMITATION THAT MATTERS

Forgetting is **silent** (no validity flag), and session adaptation (S_t updates) ≠ durable learning — fast-to-slow consolidation is open. Toy uses hash collisions; real interference is superposition along shared key directions — same shape, simpler mechanism, disclosed. Next: BDH App.E toy code, HOLA §2–3, Equations of Reasoning blog.

Sources: [1] Kosowski et al. arXiv:2509.26507 (BDH, 2025) [2] Engdahl et al. arXiv:2608.09888 (BDH-CQ, 2026) [3] Cui arXiv:2607.02303 (HOLA, 2026) [4] Gu & Dao Mamba (2024). Toy = team-written, MIT, not official BDH. AI-assisted (Muse Spark); team defends all of it. Cloud 9 · IIT KGP.